maximum-likelihood tree (IQ-TREE, ultrafast bootstraps >= 70 shown); red: appendicularians, orange: other tunicates, blue: vertebrates, purple: curated transcripts, grey: outgroup

length (aa)
Fbor_GPX.10.selenocysteine 167
Fbor_GPX.12.selenocysteine 190
Fbor_GPX.11.selenocysteine 191
Fbor_GPX.17.cysteine 109
Fbor_GPX.4.selenocysteine 225
Fbor_GPX.7.selenocysteine 179
Fbor_GPX.5.selenocysteine 170
Ostreococcus_tauri_gi_116059366_emb_CAL55073. 214
Xenopus_Silurana_tropicalis_gi_221307511_ref_ 195
Taeniopygia_guttata_gi_211971089_ref_NP_00113 197
Mmus_GPX.6.selenocysteine 201
Rattus_norvegicus_gi_9247190_gb_AAB95647.2 201
GPX1.Sus_scrofa_GPX1.Sus_scrofa 206
Molluscum_contagiosum_virus_gi_19744909_ref_N 220
Human_GPX1_NP_000572.2 203
gi_190359363_sp_Q0EF99.2_GPX1_CALJA_gi_190359 201
Macrobrachium_rosenbergii_gi_222875570_gb_ACM 186
Blan_GPX.1.selenocysteine 191
Bflo_GPX.2.selenocysteine 191
Oncorhynchus_mykiss_gi_11095769_gb_AAG30013.1 141
Drer_GPX.6.selenocysteine 190
Thunnus_maccoyii_gi_133779723_gb_ABO38817.1 188
Hypophthalmichthys_molitrix_gi_156571766_gb_A 142
Drer_GPX.2.selenocysteine 192
GPX1.Danio_rerio_GPX1.Danio_rerio 191
Blan_GPX.18.cysteine 229
Bflo_GPX.16.cysteine 227
Blan_GPX.6.serine 263
Bflo_GPX.6.serine 263
Drer_GPX.8.selenocysteine 191
Xenopus_tropicalis_gi_166796161_gb_AAI59037.1 146
GPX2.Gallus_gallus_GPX2.Gallus_gallus 191
GPX2.Sus_scrofa_GPX2.Sus_scrofa 190
Human_GPX2_NP_002074.2 190
Mmus_GPX.8.selenocysteine 190
gi_172046601_sp_Q9JHC0.3_GPX2_MOUSE_gi_172046 190
gi_72163529_ref_XP_792563.1_gi_72163529_ref_X 203
Hydra_magnipapillata_gi_221131943_ref_XP_0021 301
Ixodes_scapularis_gi_215502580_gb_EEC12074.1 123
Pediculus_humanus_corporis_gi_212511626_gb_EE 131
Drer_GPX.22.selenocysteine 194
Metapenaeus_ensis_gi_171189511_gb_ACB42236.1 174
Blan_GPX.13.selenocysteine 199
Bflo_GPX.13.selenocysteine 200
Bflo_GPX.9.selenocysteine 199
Blan_GPX.8.selenocysteine 202
Bflo_GPX.22.serine 178
GPX3.Salmo_salar_GPX3.Salmo_salar 229
Drer_GPX.16.selenocysteine 222
Xenopus_tropicalis_gi_38174689_gb_AAH61262.1 157
Mmus_GPX.14.selenocysteine 248
GPX3.Mus_musculus_GPX3.Mus_musculus 218
Human_GPX3_NP_001316719.1 235
GPX3.Homo_sapiens_GPX3.Homo_sapiens 226
Sus_scrofa_gi_212549567_ref_NP_001131079.1 221
Human_GPX6_NP_874360.1 221
Mmus_GPX.26.cysteine 233
Mus_musculus_gi_148700720_gb_EDL32667.1 233
Mmus_GPX.20.cysteine 237
gi_121671_sp_P21765.2_GPX5_MOUSE_gi_121671_sp 221
Human_GPX5_NP_001500.1 221
gi_116667974_pdb_2I3Y_A_gi_116667974_pdb_2I3Y 215
Sus_scrofa_gi_47523090_ref_NP_999051.1 219
Canis_lupus_familiaris_gi_50978980_ref_NP_001 221
Unio_tumidus_gi_111572533_gb_ABH10623.1 232
Corbicula_fluminea_gi_146395067_gb_ABQ24217.1 211
Dreissena_polymorpha_gi_145559053_gb_ABP73388 243
Hror_GPX.2.selenocysteine 136
Scla_GPX.13.selenocysteine 135
Bsch_GPX.3.selenocysteine 134
Olon_GPX.26.unaligned 105
Tdem_GPX.5.selenocysteine 133
Pege_GPX.6.selenocysteine 193
Pmam_GPX.4.selenocysteine 137
Csav_GPX.3.selenocysteine 142
Crob_GPX.11.selenocysteine 133
Cint_GPX.3.selenocysteine 134
Crassostrea_gigas_gi_152002384_gb_ABS19600.1 244
Csav_GPX.7.selenocysteine 162
Cint_GPX.9.selenocysteine 183
Cint_GPX.12.selenocysteine 163
Crob_GPX.5.selenocysteine 165
Ciona_intestinalis_gi_198424199_ref_XP_002121 136
Pmam_GPX.7.selenocysteine 163
Pmam_GPX.10.selenocysteine 159
Csav_GPX.10.unaligned 67
Csav_GPX.11.selenocysteine 119
Crob_GPX.7.selenocysteine 167
Cint_GPX.14.selenocysteine 159
gi_1708061_sp_P52033.1_GPXC_DIRIM_gi_1708061_ 221
gi_544436_sp_P35666.1_GPXC_WUCBA_gi_544436_sp 223
Caenorhabditis_briggsae_AF16_gi_157767532_ref 140
Haemonchus_contortus_gi_47499102_gb_AAT28332. 207
Globodera_rostochiensis_gi_21739125_emb_CAD38 250
Bsch_GPX.7.selenocysteine 179
Scla_GPX.9.selenocysteine 171
Scla_GPX.8.selenocysteine 167
Bsch_GPX.9.selenocysteine 162
Hror_GPX.7.selenocysteine 257
Scla_GPX.18.unaligned 111
Hror_GPX.4.unaligned 128
Scla_GPX.17.selenocysteine 167
Scla_GPX.16.selenocysteine 174
Bsch_GPX.13.selenocysteine 119
Bsch_GPX.10.unaligned 83
Olon_GPX.20.cysteine 105
Arabidopsis_thaliana_gi_15225103_ref_NP_18071 169
Rheum_australe_gi_197312911_gb_ACH63236.1 244
Ovan_GPX.28.cysteine 82
Olon_GPX.24.cysteine 78
Fbor_GPX.13.cysteine 189
Crob_GPX.8.cysteine 110
Cint_GPX.8.cysteine 137
Blan_GPX.3.cysteine 130
Bflo_GPX.3.cysteine 98
Stho_GPX.17.cysteine 135
Tdem_GPX.10.cysteine 106
Pege_GPX.8.cysteine 132
Hror_GPX.8.cysteine 162
Scla_GPX.1.cysteine 140
Bsch_GPX.1.cysteine 162
Pmam_GPX.12.cysteine 137
Drer_GPX.9.cysteine 149
Human_GPX7_NP_056511.2 187
Mmus_GPX.11.cysteine 150
Drer_GPX.11.serine 205
Human_GPX8_NP_001008398.2 209
Mmus_GPX.19.cysteine 107
Fbor_GPX.22.selenocysteine 152
Fbor_GPX.21.cysteine 59
Isaria_tenuipes_gi_210063459_gb_ACJ06531.1 170
Fbor_GPX.6.cysteine 92
Taiwanofungus_camphoratus_gi_68165856_gb_AAY8 159
Candida_albicans_SC5314_gi_68483467_ref_XP_71 163
Fbor_GPX.15.unaligned 72
Shewanella_sediminis_HAW-EB3_gi_157376430_ref 161
Bacillus_cereus_ATCC_14579_gi_30020250_ref_NP 160
Shewanella_amazonensis_SB2B_gi_119774217_ref_ 164
Fbor_GPX.16.cysteine 65
gamma_proteobacterium_NOR51-B_gi_219680407_gb 161
Sphingomonas_wittichii_RW1_gi_148556976_ref_Y 162
Rhodopseudomonas_palustris_BisB5_gi_91977935_ 158
Bradyrhizobium_japonicum_USDA_110_gi_27381843 158
Bradyrhizobium_sp._BTAi1_gi_148257534_ref_YP_ 162
Fbor_GPX.1.cysteine 163
Fbor_GPX.14.cysteine 77
Salmonella_enterica_subsp._enterica_serovar_N 183
Bordetella_avium_197N_gi_187478327_ref_YP_786 164
Bordetella_parapertussis_12822_gi_33597467_re 166
Delftia_acidovorans_SPH-1_gi_160896338_ref_YP 168
Diaphorobacter_sp._TPSY_gi_222109412_ref_YP_0 161
Acidovorax_avenae_subsp._citrulli_AAC00-1_gi_ 164
Fbor_GPX.28.cysteine 66
Fbor_GPX.24.unaligned 73
Fbor_GPX.8.cysteine 198
Fbor_GPX.20.cysteine 192
Fbor_GPX.19.cysteine 192
Fbor_GPX.25.unaligned 87
Fbor_GPX.18.cysteine 128
Fbor_GPX.3.unaligned 122
Alcanivorax_borkumensis_SK2_gi_110835499_ref_ 158
Clostridium_beijerinckii_NCIMB_8052_gi_150015 159
gamma_proteobacterium_HTCC5015_gi_198262109_g 160
Enterobacter_sp._638_gi_146312342_ref_YP_0011 160
Cellvibrio_japonicus_Ueda107_gi_192362439_ref 171
Alteromonadales_bacterium_TW-7_gi_119471461_r 165
Fbor_GPX.2.cysteine 158
Bsty_GPX.1.cysteine 158
Stho_GPX.8.cysteine 190
Stho_GPX.2.selenocysteine 179
Tdem_GPX.1.selenocysteine 176
Tdem_GPX.11.selenocysteine 157
Stho_GPX.1.selenocysteine 156
Pmam_GPX.6.selenocysteine 200
Stho_GPX.7.selenocysteine 187
Tdem_GPX.8.selenocysteine 199
Pege_GPX.3.selenocysteine 195
Pege_GPX.4.selenocysteine 183
Tdem_GPX.3.selenocysteine 217
Pege_GPX.2.selenocysteine 195
Pege_GPX.1.selenocysteine 169
Clonorchis_sinensis_gi_117956208_gb_ABK58680. 181
Csav_GPX.5.selenocysteine 204
Crob_GPX.2.selenocysteine 197
Cint_GPX.5.selenocysteine 197
Pmam_GPX.2.selenocysteine 173
Pmam_GPX.1.selenocysteine 203
Csav_GPX.1.selenocysteine 203
Crob_GPX.1.selenocysteine 204
Cint_GPX.1.selenocysteine 204
Tdem_GPX.7.selenocysteine 204
Tdem_GPX.2.selenocysteine 163
Oalb_GPX.26.selenocysteine 91
Oalb_GPX.6.unaligned 119
Ovan_GPX.24.unaligned 85
Ovan_GPX.4.selenocysteine 134
Ovan_GPX.17.unaligned 125
Ovan_GPX.8.unaligned 87
Ovan_GPX.3.selenocysteine 92
Oalb_GPX.2.selenocysteine 258
OKI_GPX.3.selenocysteine 168
OdB3_GPX.5.selenocysteine 154
OSKA_GPX.6.selenocysteine 151
Mery_GPX.2.selenocysteine 160
Bsty_GPX.3.selenocysteine 150
Olon_GPX.12.unaligned 122
Olon_GPX.30.unaligned 66
Olon_GPX.3.selenocysteine 187
Ovan_GPX.5.selenocysteine 128
Ovan_GPX.1.selenocysteine 196
Oalb_GPX.4.selenocysteine 173
Oalb_GPX.3.selenocysteine 139
OKI_GPX.1.selenocysteine 137
OSKA_GPX.9.selenocysteine 169
OdB3_GPX.2.selenocysteine 146
OdB3_GPX.1.selenocysteine 146
Hror_GPX.10.selenocysteine 201
Scla_GPX.21.selenocysteine 161
Bsch_GPX.14.selenocysteine 134
Bsch_GPX.19.selenocysteine 90
Bsch_GPX.20.selenocysteine 178
Bsch_GPX.17.selenocysteine 168
Blan_GPX.22.selenocysteine 245
Bflo_GPX.25.selenocysteine 181
Blan_GPX.16.cysteine 225
Bflo_GPX.20.cysteine 137
GPX4.Salmo_salar_GPX4.Salmo_salar 193
Drer_GPX.18.selenocysteine 170
Drer_GPX.19.selenocysteine 186
GPX4d.Salmo_salar_GPX4d.Salmo_salar 199
GPX4.Gallus_gallus_GPX4.Gallus_gallus 192
GPX4.Sus_scrofa_GPX4.Sus_scrofa 197
Human_GPX4_NP_001034937.1 234
GPX4.Homo_sapiens_GPX4.Homo_sapiens 197
Mmus_GPX.3.selenocysteine 197
Mmus_GPX.4.selenocysteine 193
Mmus_GPX.1.selenocysteine 184
Olon_GPX.1.selenocysteine 164
Olon_GPX.14.selenocysteine 142
Mery_GPX.3.selenocysteine 138
Bsty_GPX.2.selenocysteine 128
OKI_GPX.7.selenocysteine 153
OdB3_GPX.7.selenocysteine 154
OSKA_GPX.3.selenocysteine 154
Oalb_GPX.15.selenocysteine 157
Ovan_GPX.9.selenocysteine 159
Ovan_GPX.26.selenocysteine 105
Ovan_GPX.15.selenocysteine 77
Oalb_GPX.12.unaligned 114
Oalb_GPX.19.selenocysteine 154
Oalb_GPX.10.selenocysteine 167
Ovan_GPX.22.selenocysteine 166
OKI_GPX.9.selenocysteine 148
OSKA_GPX.12.selenocysteine 150
OSKA_GPX.4.unaligned 111
Olon_GPX.5.selenocysteine 140
Olon_GPX.22.selenocysteine 62
Olon_GPX.23.unaligned 92
Olon_GPX.9.selenocysteine 142
Mery_GPX.6.unaligned 100
Mery_GPX.4.unaligned 110
Mery_GPX.10.selenocysteine 73
Bsty_GPX.5.selenocysteine 124
Olon_GPX.2.selenocysteine 161
Olon_GPX.17.unaligned 91
Olon_GPX.16.unaligned 91
Olon_GPX.18.selenocysteine 74
Olon_GPX.7.selenocysteine 161
Olon_GPX.11.selenocysteine 144
Olon_GPX.4.selenocysteine 207
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
branch length